
Translate Needs To Integration Requirements

Given a use case and business process, evaluate system and process constraints.

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Table of Contents



Business Scenario

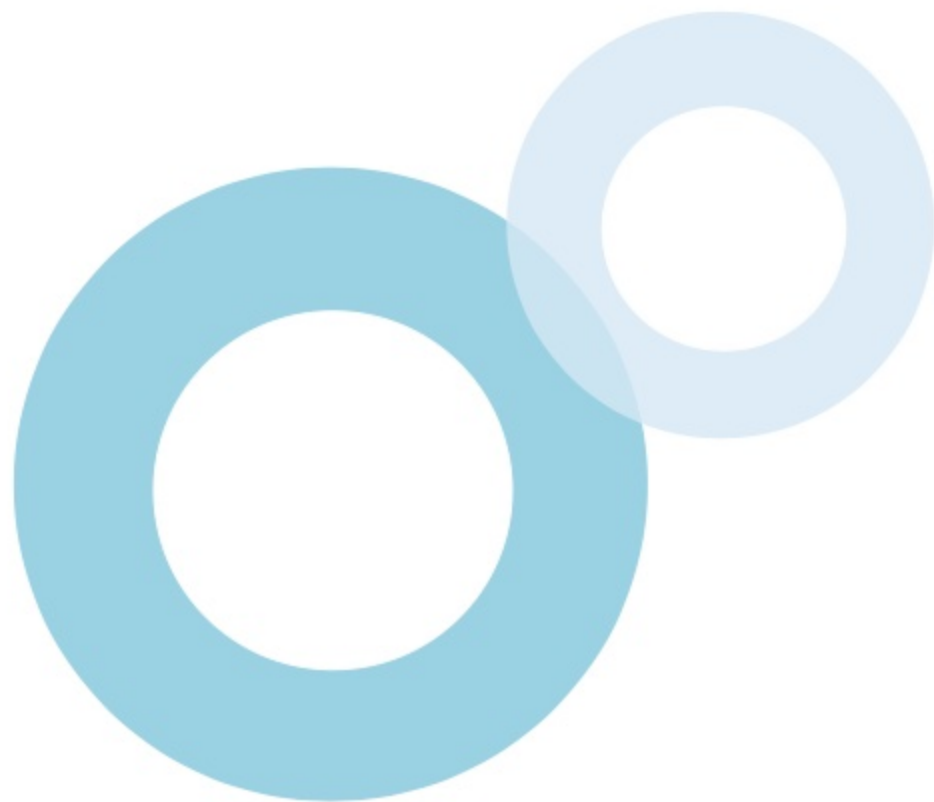


Requirements & Solutions



After studying this topic, you should be able to:

-  Define system constraints and identify how they affect integration solution design.
-  Define process constraints and identify how they affect integration solution design.
-  Evaluate system and process constraints for the given use cases and business processes.



Introduction

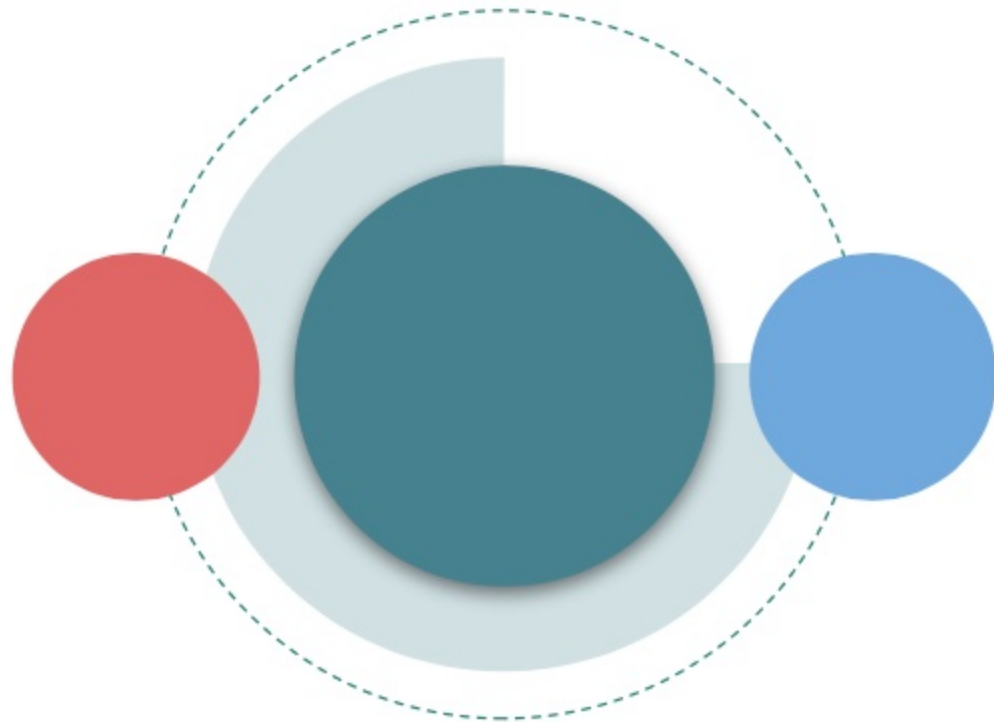
An integration design may be affected by system and/or process constraints. **System constraints** are **limitations or restrictions within Salesforce or a remote system** that necessitate the implementation of an alternative. On the other hand, **process constraints** are **limitations or restrictions in a business process** that necessitate the implementation of an alternative approach. For example, an integration solution design may be affected by the limited number of API calls per day or the lack of support for REST API by the remote system.



Overview

SYSTEM CONSTRAINTS

System constraints in an integration design refer to **limitations or restrictions within Salesforce or a remote system** that necessitate the implementation of an alternative.



PROCESS CONSTRAINTS

Process constraints in an integration design refer to **limitations or restrictions in a business process** that necessitate the implementation of an alternative approach.

System Constraints

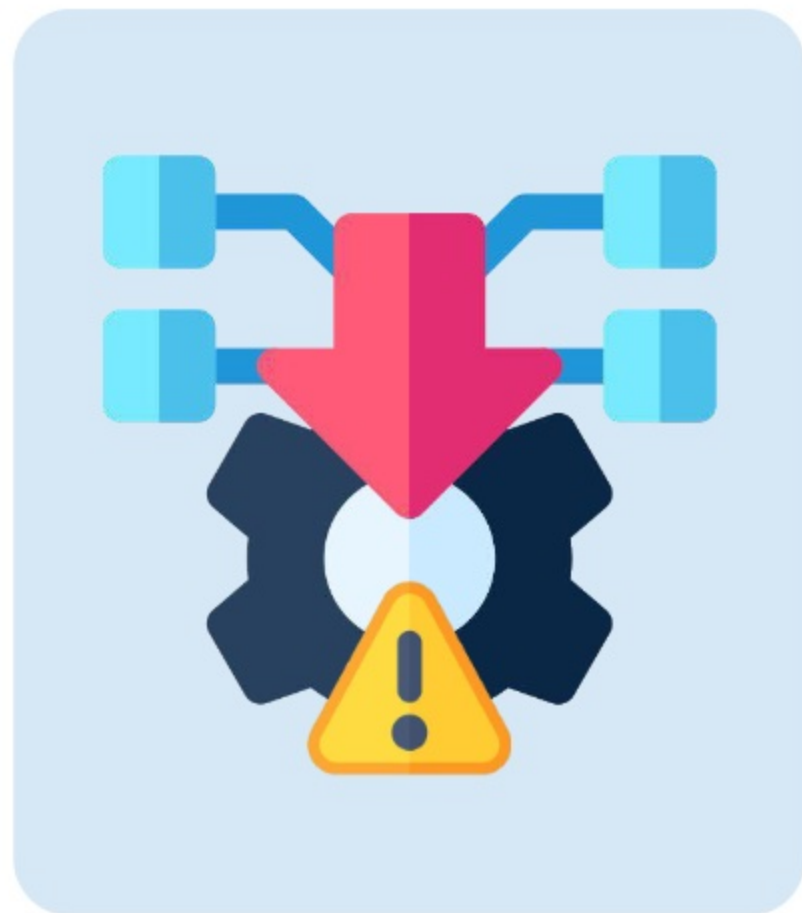
System constraints in an integration design refer to **limitations or restrictions** within **Salesforce or a remote system** that necessitate the implementation of an alternative approach.

❖ SALESFORCE CONSTRAINTS

Constraints in Salesforce may refer to **platform limitations** or **governor limits** that affect the required integration solution. For example, the number of inbound API calls per day is limited, which may affect inbound REST or SOAP API calls from a remote system, making it necessary to choose an alternative solution.

❖ REMOTE SYSTEM CONSTRAINTS

A remote system, such as an ERP system, may have certain **limitations, such as lack of support for a required feature or standard**. For example, if an ERP system does not support a RESTful service for a requirement, then External Services cannot be used.



Process Constraints

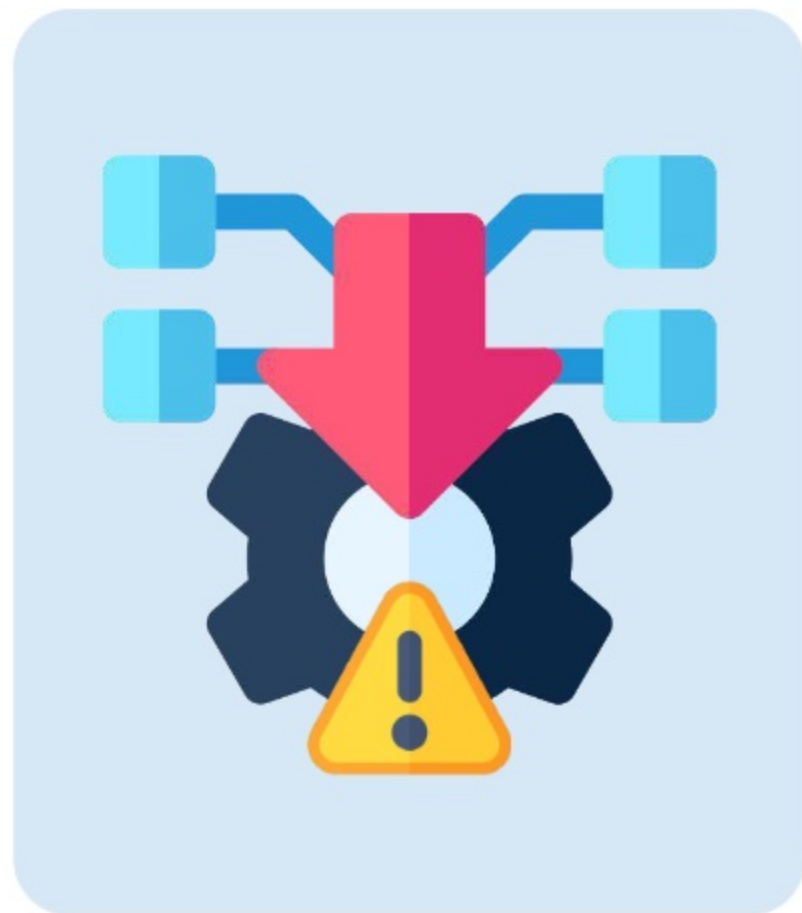
Process constraints in an integration design refer to **limitations or restrictions** in a **business process** that necessitate the implementation of an alternative approach.

❖ ACTION CONSTRAINTS

An **action required by a particular user** in a business process may make it necessary to implement a specific approach in an integration design. For example, if a field update by a user must send opportunity data to an external system, then a record-triggered flow with an asynchronous path can be used for the business requirement.

❖ COMPONENT CONSTRAINTS

A business process may require the use of a particular **component which can restrict the integration solution design**. For example, if a user must use a quick action on a record page to send data to a remote system, then the quick action can invoke a Lightning component or flow that makes a callout to the remote system for the business requirement.



Learn More



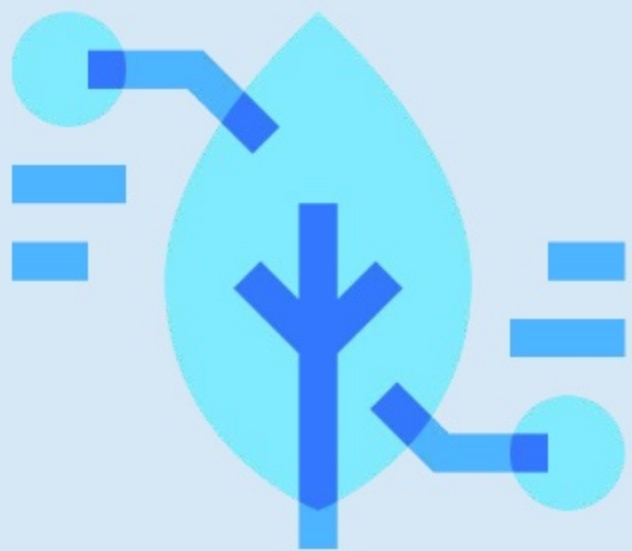
[Integration Patterns and Practices](#)

Business Scenario

Table of Contents 

Business Scenario

Read the business scenario below and determine the most appropriate solutions for the company's requirements.



Cosmic Harvest is a company in the US that mainly sells organic food items through its online store. Its sales department uses Salesforce, but the company also uses an ERP system, an SCM (Supply Chain Management) system, and a WFM (Workforce Management) system.

The company has an upcoming integration project for which the solution architect needs to analyze certain use cases and evaluate the system and process constraints that could affect the integration solutions.

Requirements & Solutions

[Table of Contents](#)



Requirement 1



SCENARIO

The company's sales reps use Salesforce for managing accounts, contacts, and opportunities. When an opportunity reaches the 'Closed Won' stage, an order must be created in the ERP system. The sales rep who updates the opportunity requires an immediate response once the order has been created. The sales rep must use a Lightning web component to update an opportunity. In addition, the ERP system does not have a RESTful service that could create an order automatically. The solution architect needs to identify how these system and/or process constraints could affect the type of integration solution that is used for the requirement.

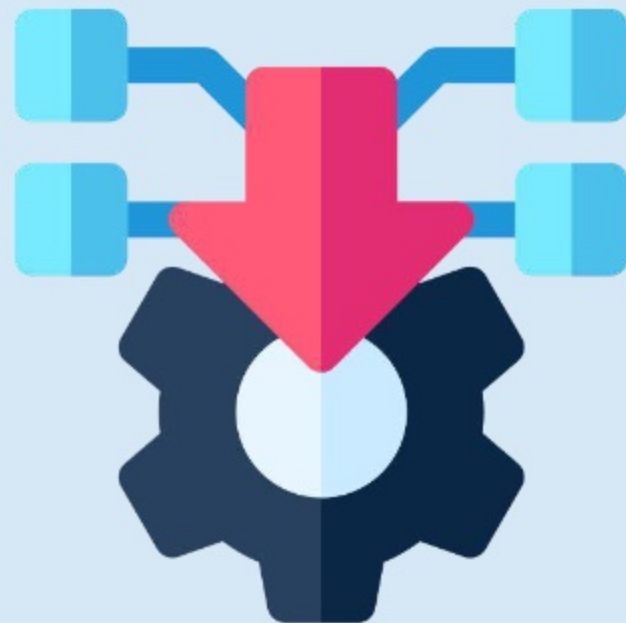


Solution 1



SOLUTION

If the ERP system does not have a RESTful service that could be used for the requirement, then using **External Services** or **Apex REST callouts** would not be possible. If a sales rep must use a Lightning web component to update an opportunity, then **Apex code** can be used to perform, if supported, a SOAP callout instead to the ERP system. A **record-triggered flow** could be used if an opportunity update would need to trigger a callout to the ERP system automatically.



Requirement 2



SCENARIO

The company would like to integrate products in Salesforce with its Supply Chain Management (SCM) system. When a new product record is created in Salesforce, the same product should be created in the SCM system. The user should not need to wait for the successful creation of the product in the SCM system. However, there are certain limitations. The SCM system does not support the use of CometD. A Salesforce user needs to use a Visualforce page to create a new product. Any related considerations need to be identified.



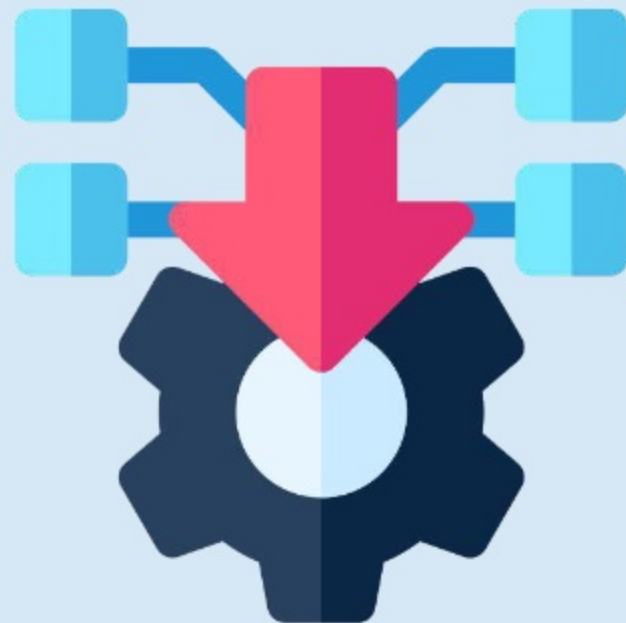
Solution 2



SOLUTION

If the Visualforce page publishes a **platform event** and the SCM system does not support **CometD**, then the **pub/sub API** can be used by the SCM to subscribe the platform event.

On the other hand, if an **Apex SOAP** or **HTTP callout** is used to initiate the record creation in the SCM, the request should be executed asynchronously since the user should not wait for a successful response from the SCM. In this case, the Apex SOAP or HTTP callout can be performed asynchronously by having the Visualforce page call a **future method** or create a **queueable job**.



Requirement 3



SCENARIO

The ERP system stores all the orders placed by customers. It was built by the company's developers several years ago. The company's sales users regularly use Salesforce, but they need to log in to the ERP system every time they need to check order information. The sales director wants to move new orders to Salesforce on a regular basis and relate them to account records to enable easier access. In addition, any existing orders in Salesforce must be updated. However, a suitable third-party ETL tool is unavailable for the required data loading operations, and the ERP system does not support making SOAP callouts. In addition, Salesforce receives API calls from multiple enterprise systems daily. Any related considerations need to be identified.



Solution 3



SOLUTION

If a suitable third-party ETL tool is unavailable, then an **ETL tool cannot** be used for extracting, transforming, and loading order data into Salesforce. If the ERP system cannot make SOAP callouts, then a **'Remote Call-In' using SOAP API cannot** be used to send order data to Salesforce. If Salesforce receives multiple API calls from other systems, then it would be important to consider whether the new integration solution would affect the **daily limit** associated with the **number of inbound API calls**. In this case, a Remote Call-In using REST can be implemented instead. To transform data so that it can be saved as a record in Salesforce, custom Apex logic can be performed by building an Apex REST service. To minimize API calls, the operation can be done on a scheduled basis and process records in batches.



Requirement 4



SCENARIO

The company's HR department uses a workforce management (WFM) system to manage and process employee data. When the information of an employee who is a Salesforce user is updated in the WFM system, the user record of the employee in Salesforce should be updated automatically. The WFM system is not designed to perform outbound calls to any system, but it can support inbound calls. The architect must identify how this would affect the integration design and suggest a solution.



Solution 4



SOLUTION

The ideal workflow is that the WFM system should initiate the update in Salesforce by performing an outbound request when an employee record is updated in its system. Due to the limitation and provided that a middleware is not available, Salesforce will have to initiate the request by performing an outbound call to the WFM using a **scheduled job**. A **scheduled Apex job** can be created to consume either a SOAP/REST web service. Or, a **scheduled-triggered flow** can be used to perform an HTTP callout or consume a REST-based web service. Optionally, to run jobs in smaller time intervals, multiple job instances in different recurring timings can be created.

